

Module 57 - DOM Manipulation

1. What is DOM Manipulation?

DOM Manipulation is the process of creating, changing, adding, replacing, or removing HTML elements and their content using JavaScript.

DOM Manipulation Methods

- **createElement()** → Creates a new HTML element.
- **innerText** → Gets or changes the visible text of an element.
- **innerHTML** → Gets or changes the HTML content inside an element.
- **textContent** → Gets or changes all text content inside an element.
- **append()** → Adds content or elements at the end.
- **appendChild()** → Adds an element as the last child.
- **prepend()** → Adds content or elements at the beginning.
- **insertBefore()** → Inserts an element before a specified child.
- **replaceChild()** → Replaces an existing child with another element.
- **removeChild()** → Removes a specified child element.
- **remove()** → Removes the selected element itself.
- **style** → Changes the CSS styles of an element.

Example Code:

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM Manipulation</title>
  <style>
    .container{
      background-color: red;
      width: 400px;
      margin: auto;
      padding: 10px;
      box-shadow: 10px 10px 3px black;
    }
  </style>
</head>
```

```
<body>

  <!-- Container holding the input, button, and list -->
  <div class="container">

    <!-- Takes input from the user -->
    <input type="text" placeholder="Enter Input">

    <!-- Calls manipulateElements() when clicked -->
    <button onclick="manipulateElements()">Click Me</button>

    <!-- Existing ordered list -->
    <ol>

      <li>item-1</li>

      <li>item-2</li>

      <li>item-3</li>

    </ol>

  </div>

  <script src="index.js"></script>
</body>
</html>
```

Index.js

```
function manipulateElements() {

  // Create li element

  let newListElement = document.createElement("li");

  // Add text

  // newListElement.innerText = "<a>Link</a>Item-4";

  // Add HTML

  // newListElement.innerHTML = "<a>Link</a>Item-5";

  // Add plain text

  // newListElement.textContent = "<a>Link</a>Item-6";

  // Select input

  let input = document.querySelector("input");

  // Get input value

  // console.log(input.value);
```

```
// Set input value as text
newListElement.innerText = input.value;

// Select ordered list
let orderList = document.querySelector("ol");

// Add at end
// orderList.append("Text Node", newListElement);

// Insert before element
// orderList.insertBefore(newListElement, orderList.children[2]);

// Replace element
// orderList.replaceChild(newListElement, orderList.children[2]);

// Remove child
// orderList.removeChild(orderList.children[0]);

// Remove element
// orderList.remove();

// Add at beginning
// orderList.prepend("Text Node", newListElement);

// Change text color
newListElement.style.color = "yellow";

// Add text shadow
newListElement.style.textShadow = "10px 10px 1px white";

// Add element at end
orderList.appendChild(newListElement);
}
```